

■ Study Session 3 — Correlation & Regression

■ Goals

- Understand correlation and multicollinearity.
- Run simple and multiple linear regression with scikit-learn.
- Interpret slope, intercept, and R^2 values.
- Visualize and evaluate predictions (Actual vs Predicted with $y = x$).

■ Steps

- 1 Load cleaned dataset (tips_cleaned.csv) and inspect columns.
- 2 Compute correlation matrix → `df.corr(numeric_only=True)`; plot a heatmap.
- 3 Simple linear regression: predict tip_usd from bill_total_usd.
- 4 Record slope, intercept, R^2 . Interpretation example: every \$10 higher bill ≈ \$1.05 higher tip.
- 5 Plot scatter with regression line (`sns.regplot`) and explain trend briefly.
- 6 Add an ELI5 markdown note for slope and R^2 .
- 7 Multiple regression: add party_size, gender, is_smoker as predictors.
- 8 Plot Actual vs Predicted tips with a red dashed $y = x$ reference line.
- 9 Save charts using `save_and_log_plot()` so plots_index.md can index them.

■ Reflection Questions

- How well does bill size alone explain tips (R^2)?
- Which added features improved R^2 the most?
- Any strong correlations that suggest multicollinearity?
- Does higher bill always mean higher tip percentage?

■ ELI5 Summary

Think of regression as a smart guesser: it learns that bigger bills usually mean bigger tips. The red dashed line on the Actual vs Predicted plot is the 'perfect guess' line—if all your dots sit on it, your model guessed every tip exactly right.